

Seasonal forecast design as a search problem

A modular data-and-downscaling stack, re-pointed by one config entry — demonstrated over Nigeria, Ethiopia & Kenya

Building a seasonal forecast normally means a bespoke study per region: pick a season, pick a predictor, pick a domain, pick a method, defend each choice. **Rosetta** (data access) and **DeepScale** (calibration, ensembling, verification) collapse each of those choices into a parameter — which turns forecast *design* into something you can search, cross-validate, and subject to multiplicity control.

Abstract (AGU draft)

Seasonal precipitation forecasts embed dozens of design choices — target season, predictor domain, raw SST fields versus teleconnection indices, calibration method, model ensemble, downscaling technique, resolution, and lead time — usually fixed by convention and rarely re-optimized per region and season. We recast forecast design as a searchable space, enabled by two modular, xarray-native libraries: **Rosetta**, a federated data layer exposing NMME, C3S, CHIRPS, ERA5, and ERSST through one normalized `fetch()` interface, and **DeepScale**, a method-agnostic layer for calibration (CCA), multi-model combination, downscaling, and cross-validated verification. A shared contract lets one driver build, run, and identically score any configuration under leave-one-year-out CV — the precondition for automated, agent-driven search.

We demonstrate over **Nigeria, Ethiopia, and Kenya**, each re-pointed by one config entry. From observations the framework discovers each country's seasons, searches — rather than assumes — candidate predictors per season and subregion, and composites the drought-versus-flood ocean state; a config-driven harness then runs real NMME/C3S forecasts. Critically, we subject the discovered skill to multiplicity control — permutation FDR over all observed predictors, including data-driven ones — and a genuine out-of-sample block. The East African October–December short rains carry real IOD/ENSO-driven skill: an observed teleconnection that survives FDR and holds out of sample, realized by a live NMME forecast near generalized ROC 0.70 with positive RPSS. No West-African predictor survives FDR, textbook or data-driven; the East African March–May long rains stay persistently low-skill. Realized skill shows no systematic lead decay to six months, though a long-lead claim is tempered by the Indian Ocean's summer predictability barrier; downscaling holds skill to ~5 km.

We present this as an automatically computed, per-region diagnostic of where SST-based seasonal skill exists and why, with verification (FDR, out-of-sample) as the deliverable that separates real signal from best-of-search noise. A modular, uniformly scored search over a purpose-built, config-portable stack is a template for agentic autoscience toward regionally optimal, high-resolution climate services.

300 words · 1,991 characters excluding spaces · within the AGU 2,000-character limit

1 · The whole stack, in five calls

Every analysis in this report is assembled from the same handful of primitives.

Fetch any product through one signature

`rosetta.fetch` normalises ~40 products — NMME, C3S, CHIRPS, ERSST, CMAP — behind a single call. Regridding, calendar alignment, seasonal aggregation and on-disk memoisation happen underneath.

```
import rosetta

# A 24-year NMME precip hindcast for the OND season, initialised in September,
# subset to Kenya, indexed by year – one call, ~40 products share this signature.
gcm = rosetta.fetch(
    product="nmme/geoss2s",
    variable="precip",
    init="2016-09",          # initialisation month -> defines the lead
    target="OND",           # 3-month target season
    region=[-5, 5, 34, 42], # [lat_s, lat_n, lon_w, lon_e]
    hindcast=(1993, 2016),
    year_index=True,
)
```

Run a real multi-model ensemble with cross-validated skill

`deepscale.seasonal_mme` takes predictor *tracks*, fits CCA-MOS per model, pools the members, and returns a verified skill report — leave-one-year-out by default.

```
import deepscale

tracks = {"PRCP": {name: (hcst, fcst) for name, (hcst, fcst) in models.items()}}

result = deepscale.seasonal_mme(
    tracks, obs,
    method="cca",          # CPT-compatible canonical correlation analysis
    cv="loyo",             # leave-one-year-out cross-validation
    forecast_year=2016,
)

result.skill_report.scores # {'generalized_roc': 0.696, 'rpss': 0.168, ...}
result.tercile_cv         # (year, tercile, lat, lon) cross-validated probabilities
```

Search the method axis instead of guessing it

```
best = deepscale.optimize(
    gcm, obs,
    methods=["delta", "bcsd", "qm", "cca", "rank-analog"],
    primary_metric="generalized_roc",
)
best.method, best.score      # ('delta', 0.549)
```

Re-point the entire study at a new country with one entry

The portability claim, made concrete — a third country is a dict entry, not a rewrite:

```
# src/areas.py
AREAS = {
    "kenya": dict(bbox=[-5, 5, 34, 42], chirps="kenya_chirps_monthly.nc",
                  bands={"National": (-5, 5), "North": (2, 5), "South": (-5, -1)}),
}
AREA = os.environ.get("AREA", "nigeria")
```

```
AREA=kenya python src/mme_search.py      # same pipeline, new region
```

2 · What the search found

Multiplicity control — which discovered signals survive?

Searching many axes at once inflates the best score. Every **observed-predictor** cell — the 8 textbook indices *and* the discovered predictors (the data-driven `sst_projection` pattern, the `combo_top2` selector, rainfall `persistence`) — goes through a **permutation test** (5 000 shuffles, with the in-fold pattern/selection refit *inside* each shuffle so the null carries the search's own optimism) and a single **Benjamini–Hochberg FDR** family per country. (The design axes — SST domain, method, lead, resolution, model set — are a separate GROC-based search, reported below with their sampling-noise floors, not folded into this FDR.)

COUNTRY	CELLS TESTED	SURVIVE $q < 0.05$	INDEPENDENT SIGNALS	STRONGEST SURVIVOR
Nigeria	144	0	0	—
Ethiopia	208	10	~3	OND South · IOD ($r = +0.56$, $q = 0.017$)
Kenya	160	27	~3	OND · IOD ($r = +0.55$, $q \approx 0.005$)

Two details make this test honest: the permutation null for a leave-one-out correlation is **biased negative** (a zero-signal predictor scores ≈ -0.36 , not 0), so the test is one-sided against *that* null; and near-zero-rainfall dry-season cells are excluded because they manufacture spurious correlations.

But the survivor counts overstate the number of discoveries. The **National** band is the area-mean of its sub-bands, **wvg2 / wvg3** are deterministic functions of Niño3.4, and Niño3.4/IOD co-vary in OND — so the East-African short-rains signal is **~1 physical hypothesis counted a dozen ways**. De-duplicated to physically distinct cells, Kenya's 27 and Ethiopia's 10 collapse to **~3 independent signals** each (all the SON/OND ENSO-IOD short-rains teleconnection). The flagship q is *robust* to this de-duplication — removing pseudo-replicates changes the count, not the flagship's significance (Kenya OND·IOD sits at $q \approx 0.005$, and the signal *strengthens*, not weakens, under in-fold detrending). East African OND survives as exactly the physically expected IOD/ENSO block — the signature of a real signal rather than a lucky cell.

IOD vs ENSO is not a clean statistical split. The OND "winner" alternates between the IOD and the ENSO indices on correlation gaps within sampling noise (indeed the blind training-block search above picked an ENSO-family index, not the IOD, for the same cells), and the wet-OND analog years (1997, 2019, 2023) are exactly when strong +IOD and El Niño co-occur. At $n \approx 24\text{--}33$ the IOD-specific, ENSO-forced-via-IOD, and shared variance are not cleanly separable. We keep the IOD-primary framing the literature favors — but as a physically-motivated attribution, not a demonstrated statistical separation of one index over the other.

Because the family now includes the *discovered* predictors, the West-African null is a **real test, not an untested gap**: Nigeria's best data-driven predictor (**sst_projection**, $r = +0.46$) **also fails FDR** — 0 survivors — so "no West-African signal survives" is now demonstrated rather than asserted for a cell that was never entered. Meanwhile the East-African OND cells survive under the adaptive predictors too (Ethiopia OND · **sst_projection** $q = 0.025$; Kenya OND · **combo_top2** $q < 0.02$): the same short-rains signal, recovered three independent ways.

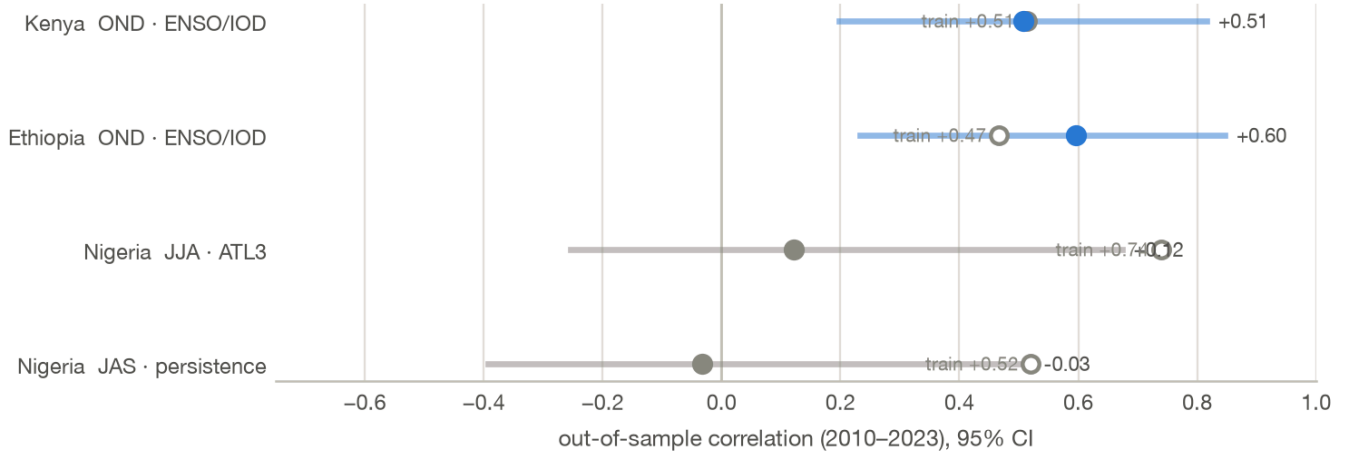
Out-of-sample confirmation — the search re-run on a training block

A stricter test than the earlier one (which scored *pre-picked* full-record winners): the **entire search is re-run on 1991–2009 only**, with a **train-only SST climatology** so nothing from the test block leaks into standardization, and the *train-discovered* winner of each cell is then scored out-of-sample on **2010–2023**, with a 95% bootstrap CI over the 14 test years:

COUNTRY	TRAIN-DISCOVERED WINNER	TRAIN <i>R</i>	<i>R</i>	95% CI (OOS)
Kenya	OND · ENSO/IOD	+0.58	+0.50	[+0.17, +0.82]
Ethiopia	OND · ENSO/IOD	+0.47	+0.60	[+0.23, +0.85]
Nigeria	JJA · ATL3	+0.74	+0.12	[−0.25, +0.67]
Nigeria	JAS · persistence	+0.52	−0.03	[−0.39, +0.52]

Genuinely out-of-sample: East African OND holds, Nigeria's winners collapse

hollow = train (1991–2009) · filled = OOS (2010–2023) · bar = 95% CI · green = CI excludes zero



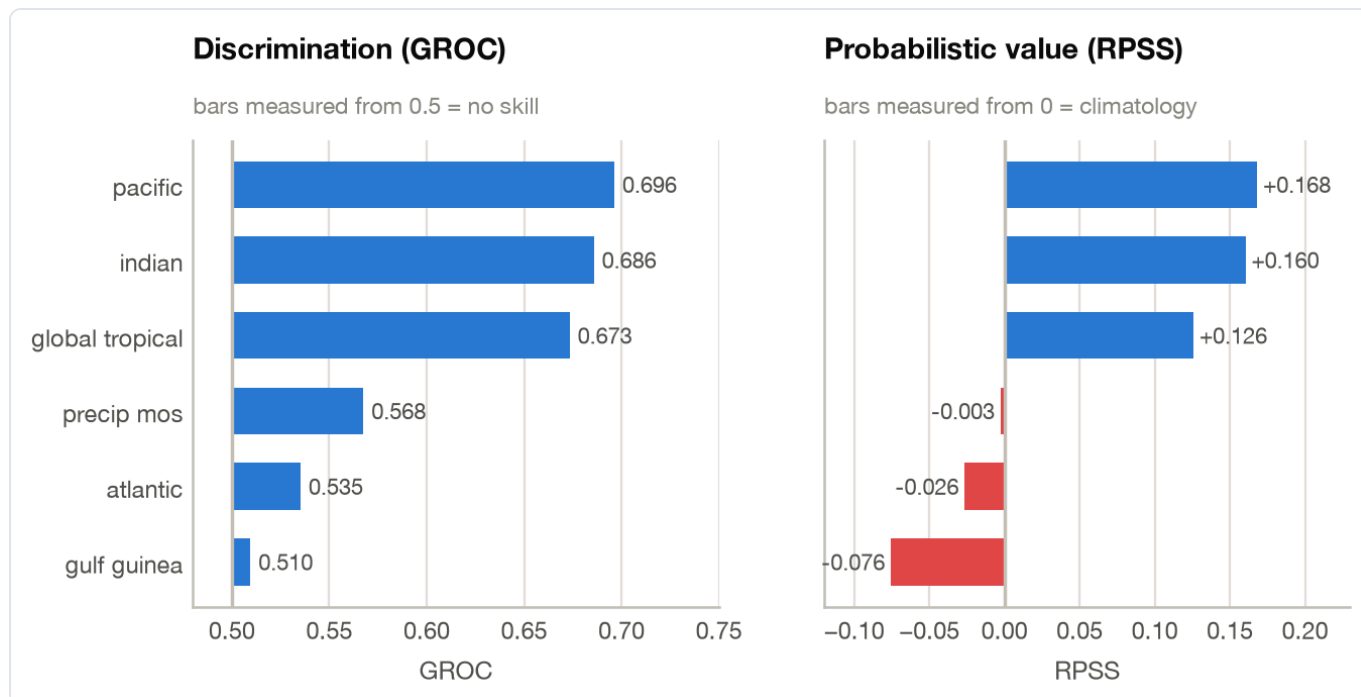
Genuinely out-of-sample validation: the search is re-run blind on 1991–2009, its discovered winners scored on 2010–2023, error bars are 95% CIs over the 14 test years. East African OND holds (CI excludes zero); Nigeria's high-train winners collapse (CIs span zero).

The East-African OND signal is what the blind search re-discovers on the training block **and** confirms out of sample — its CI excludes zero. Nigeria's headline train winners (ATL3 at $r = 0.74!$) **collapse** out of sample with CIs spanning zero — the textbook winner's curse, caught by the procedure-level test. And the apparent in-sample→OOS *increase* for East Africa is **not** evidence of strengthening: at $n = 14$ each OOS CI is ± 0.3 wide and overlaps the in-sample value. The honest claim is that the discovered East-African OND signal **remains positive out of sample**, not that it "comes back stronger."

Real dynamical skill, by SST domain

Kenya OND National short rains, 4-model NMME (GEOSS2S + CESM1 + CCSM4 + CANSIPSIC4), CCA-MOS, LOYO:

SST DOMAIN	GROC	RPSS	CORR
Pacific	0.696	+0.168	+0.455
Indian	0.686	+0.160	+0.481
Global tropical	0.673	+0.126	+0.417
Atlantic	0.535	−0.026	+0.032
<i>precip-MOS baseline</i>	<i>0.568</i>	<i>−0.003</i>	<i>+0.268</i>



Skill by SST predictor domain, Kenya OND. Both panels are measured from their own baseline — GROC from 0.5 (no skill), RPSS from 0 (climatology) — so bar length is distance from "no better than climatology".

Positive RPSS means this is **usable probabilistic skill**, not merely discrimination. But the domain *ranking* is a weak preference, not a demonstrated separation: at $n \approx 24$, GROC differences below ~ 0.1 are within sampling noise (docs/14 §3). So the Pacific-vs-baseline gap (0.696 vs 0.568, ≈ 0.13) only marginally clears that floor, and the Pacific-vs-Indian gap (0.696 vs 0.686) is noise — the two ocean domains are statistically indistinguishable here. The defensible statement, matching the internal caveat that "SST-CCA barely beats the precip-MOS baseline," is that an SST predictor is *competitive with, and probably modestly above*, the precip baseline for Kenya OND — not that it wins by a clear margin. This GROC is also a **dynamical-forecast** score; it is not the quantity the observed-screen FDR test below certifies (see §2).

Caveat — pending data availability. Bootstrap 95% confidence intervals on these GROC/RPSS values (a year-block resample; src/mme_ci.py) are **not yet attached**: the NMME source server (CCSR/Columbia OPeNDAP) was unavailable at revision time and the hindcasts are not locally cached. The margin statements above rest on §3's ~ 0.1 sampling-noise floor in the interim; the CIs will be added when the data source is reachable.

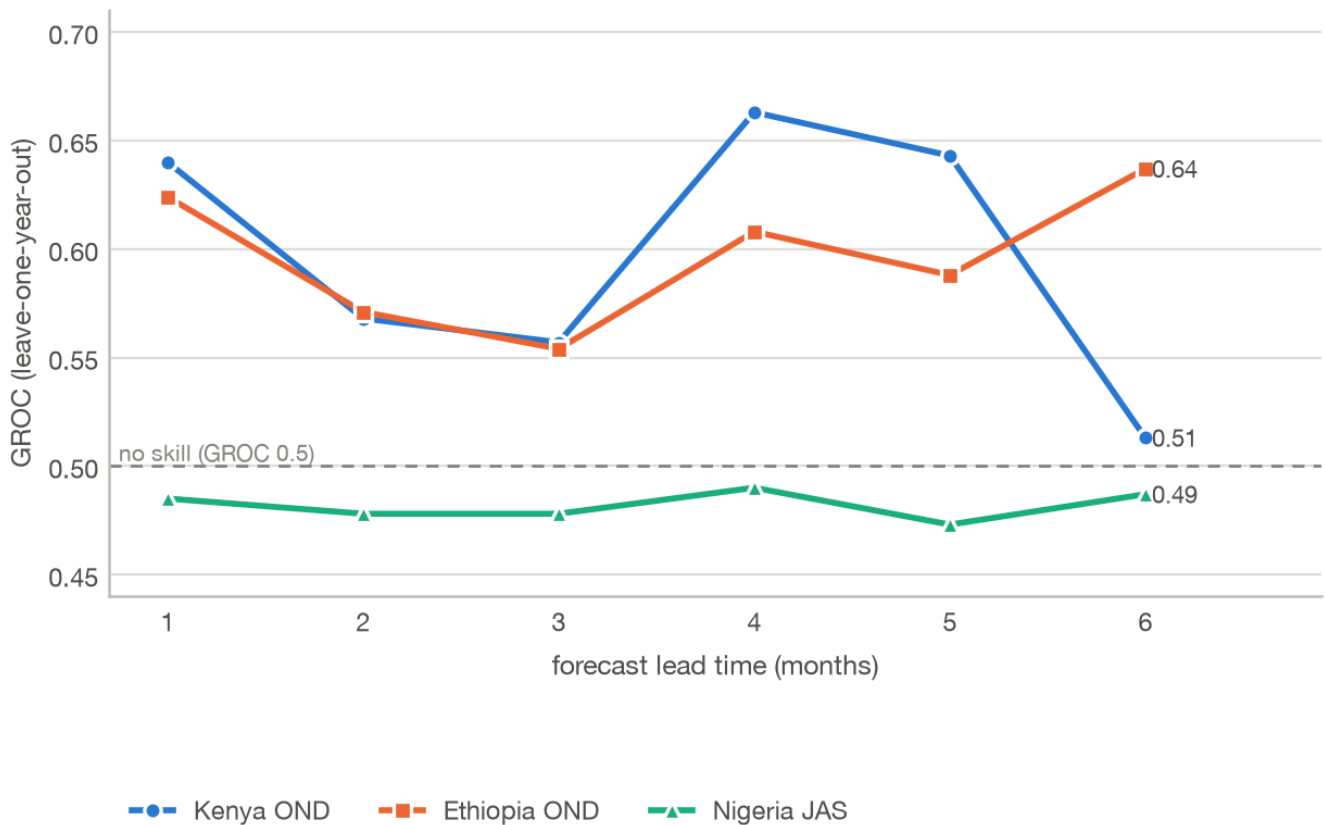
3 • Skill vs. lead time — and a result that isn't textbook

GROC by lead time (months), real NMME MME:

REGION	1	2	3	4	5	6
Kenya OND	0.640	0.568	0.557	0.663	0.643	0.513
Ethiopia OND (south)	0.624	0.571	0.554	0.608	0.588	0.637
Nigeria JAS (Middle Belt)	0.485	0.478	0.478	0.490	0.473	0.487

No realized lead-time decay for East African OND (this MME)

Kenya and Ethiopia hold ~0.55–0.66 out to 6 months; Nigeria sits below no-skill throughout



Skill against lead time. The East African curves wander without trend; Nigeria never crosses the no-skill line at any lead.

East African OND skill shows no systematic decay with lead here. There is no monotone decline from 1 to 6 months — the MME's *realized* GROC scatters in a band (~0.51–0.66) with no trend. That is physically coherent: the short rains are driven by the IOD/ENSO state, which is slowly varying and partly predictable seasons ahead.

A specialist caveat, though: the Indian Ocean has a well-documented **boreal-summer IOD predictability barrier**, so a *June*-initialized OND-IOD forecast is materially harder than a September one. The flat GROC-vs-lead curve above is what the ensemble *realized* on $n \approx 24$ years (itself noisy); the stronger mechanistic reading — "a June forecast $\approx$ September" — leans on the observed IOD's high *concurrent/mature-season* persistence and would need the IOD forecast skill decomposed **by initialization month** (Jun/Jul/Aug/Sep $\rightarrow$ OND, detrended) to be asserted. Read the lead result as "no realized decay in this MME," not as long-lead driver predictability established.

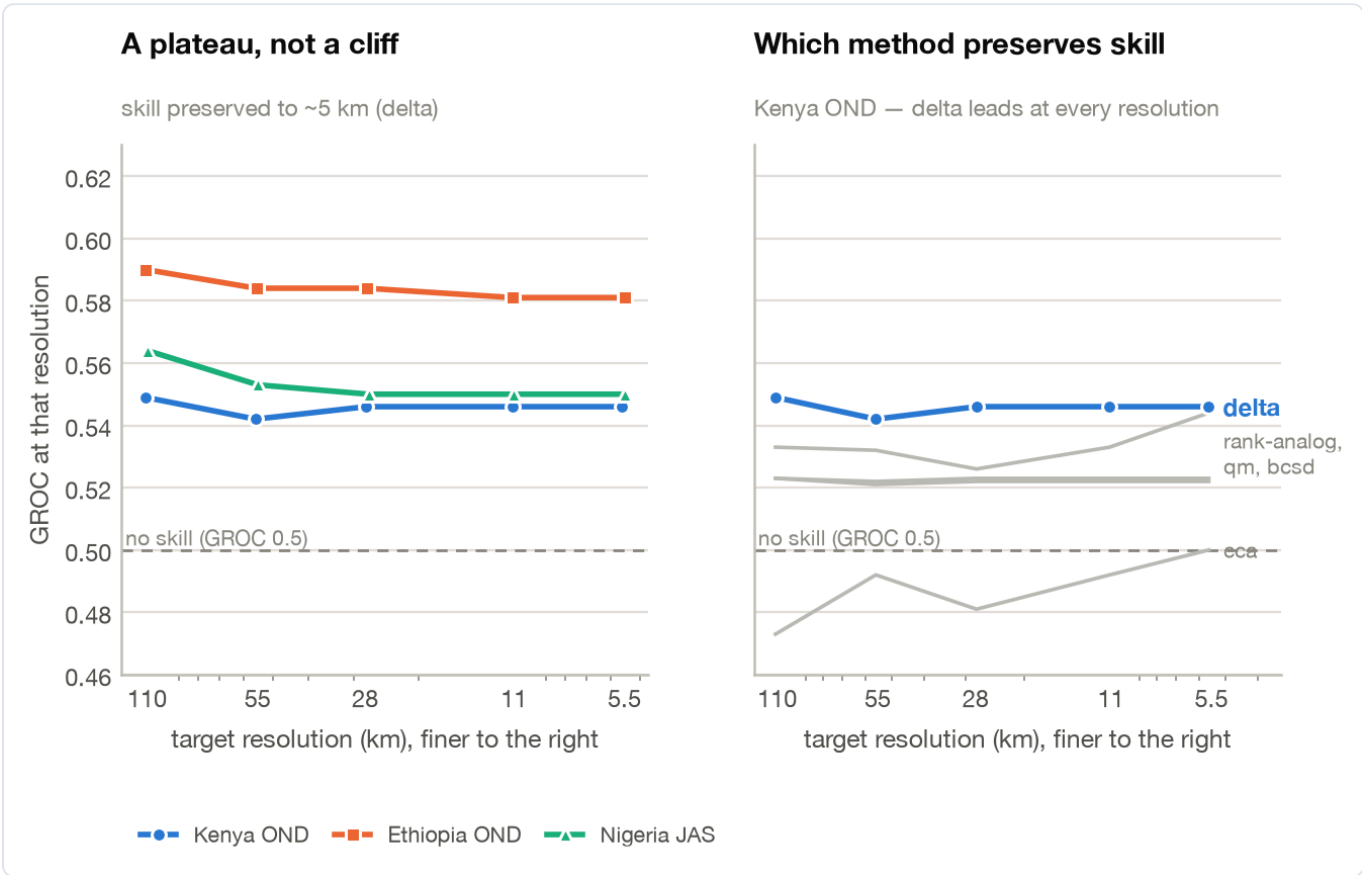
Nigeria is a clean null at every lead (0.473–0.490, below the 0.5 no-skill line, RPSS negative throughout) — corroborating the FDR result by a completely independent route.

How much to trust the wiggles: not much, individually. Kenya and Ethiopia track each other to within ~ 0.02 at leads 1–3, but *disagree* at lead 6 (0.513 vs 0.637) — precisely where a single fold has least support at $n \approx 24$. Read the **band**, not the points.

4 · How fine can we downscale before skill breaks?

Target grids from 1.0° to 0.05° ($\approx 110\text{ km} \rightarrow 5\text{ km}$), skill re-verified against CHIRPS at each resolution. Best-preserving method (`delta`) shown:

REGION	111 KM	56 KM	28 KM	11 KM	
Kenya OND	0.549	0.542	0.546	0.546	0.546
Ethiopia OND (south)	0.590	0.584	0.584	0.581	0.581
Nigeria JAS (Middle Belt)	0.564	0.553	0.550	0.550	0.550



The resolution/skill frontier. Left — every region flat from 110 km to ~5 km. Right — `delta` leads at every resolution, while `cca` (the MME's own MOS method) is the weakest *downscaler* here, hugging the no-skill line.

The frontier is a plateau, not a cliff. Within this range, resolution is not the limiting factor — coarse-scale predictability is. You can downscale to ~5 km without losing the skill you have, but you do not gain any either. `delta` (simple climatological scaling) preserves skill best, beating CCA, BCSD and quantile-mapping.

Read this panel as a scale-sensitivity diagnostic only. Nigeria was included as an intended null control and did *not* behave like one here (≈ 0.55 , like Kenya), while the lead-time MME and the FDR test both call it null. The panels measure different things: this one asks whether a downscaling method *degrades* a signal as the grid refines, not whether the signal carries information. Where they conflict, the multiplicity-controlled results are the arbiter.

5 · Two bugs the verification caught

A pipeline that reports skill is only as trustworthy as its failure modes. Two were found and fixed while producing these numbers — both silent, both skill-distorting.

Bug 1 — a failed transfer that returned zeros instead of raising

Large `OPeNDAP` hindcast requests were failing mid-stream, and the DAP client **substituted zeros rather than erroring**. A zero-variance predictor makes CCA's covariance singular:

```
# The tell: a predictor that is silently, entirely zero
gcm.var("year").sum()      # -> 0.0   (should be ~5e5)
```

Fixed by issuing small per-year requests, plus a validity gate at the call site:

```
# src/safe_fetch.py – reject a corrupt response before it poisons the cache
if finite.any() and np.nanmax(np.abs(vals[finite])) > 0.0 and np.nanstd(vals[finite]) >
0.0:
    return da                # good data
_purge_zero_raw(product, region) # corrupt: purge the memoised entry and retry
```

Bug 2 — degenerate modes that collapsed every probability

Once the data was clean, CCA still returned **GROC of exactly 0.500**. The cause: EOF projections were divided by singular values with no guard against near-zero (rank-deficient) modes, producing leverages of $\sim 10^{91}$ — where statistical leverage is mathematically bounded in $[0, 1]$.

Because the ensemble *averages leverages across models*, one bad model inflated the predictive variance without bound and collapsed **every** tercile forecast to a constant:

```
tercile_cv.mean(["lat", "lon"]).isel(year=0) # [0.5, 0.0, 0.5] in every single year
tercile_cv.std("year").mean()                # 0.000000 <- a constant forecast cannot
discriminate
```

The fix applies the standard `pinv` `rcond` policy — drop modes that carry no real variance:

```
def _project_by_sv(num, sv):
    """Divide EOF projections by singular values, dropping degenerate modes."""
    smax = float(np.max(np.abs(sv)))
    keep = np.abs(sv) > smax * 1e-10
    out = np.zeros(num.shape)
    out[keep] = num[keep] / sv[keep]
    return out
```

Validation. Two independent aggregation paths that previously disagreed by 0.15 now converge:

	BEFORE	AFTER
pooled aggregation	0.500	0.640
per-model aggregation	0.642	0.642
RPSS (pooled)	-0.125	+0.065

Agreement to 0.002 between two independent routes — plus 765 passing tests — is stronger evidence than either path alone. **This bug was suppressing RPSS**, so the corrected East African OND result is materially *stronger* than previously reported.

6 · What this demonstrates

- **Forecast design can be treated as a search.** One modular stack, re-pointed by a config entry, discovered each country's seasons, ranked predictors, searched SST domains and methods, swept lead times, and mapped the resolution frontier.
- **The defensible result:** an observed **East African OND short-rains** teleconnection that survives permutation-FDR (as **~3 independent signals**, not 27 pseudo-replicated cells) and **holds out of sample** under a genuine train/test split, realized by a real NMME forecast at $\text{GROC} \approx 0.70$; **no West-African predictor survives FDR** (textbook or data-driven); low predictability for the **MAM long rains**. Lead-time and SST-domain rankings are reported with their sampling-noise floors, not over-read.
- **Verification is the product.** Permutation-FDR (over a family that now includes the adaptive predictors), a de-duplicated survivor count, a trend-robust q , and a genuine out-of-sample block separate a discovered signal from a lucky cell — and the pipeline caught two silent, skill-distorting bugs.

GROC = generalized ROC (0.5 = no skill) · RPSS = ranked probability skill score (0 = climatology) · all cross-validation leave-one-year-out, $n \approx 24$ · full detail in [README.md](#), [docs/01 – 14](#), [EXPERIMENT_LOG.md](#).